

Why Did American Student Achievement Decline?

Verifying the post-2013 fall and testing its explanations

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Motivation: the largest sustained decline on record

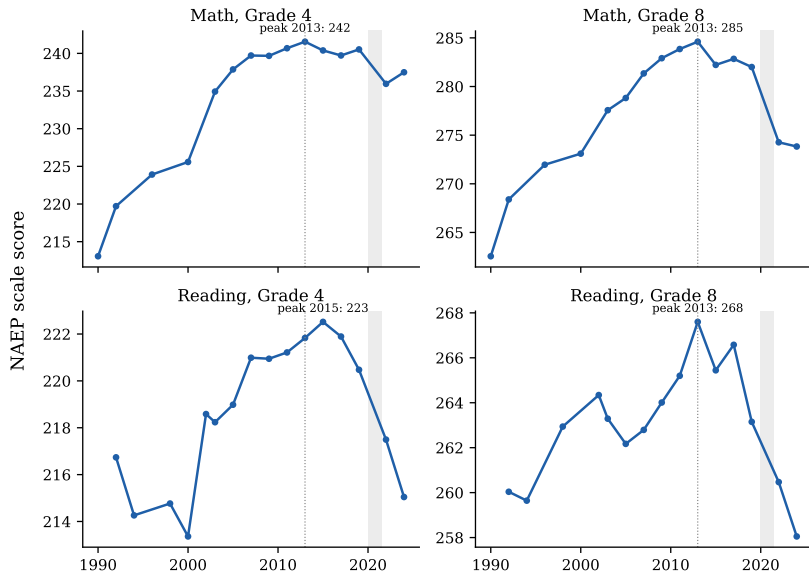
- ▶ U.S. achievement peaked around **2013** and has fallen since — seven years *before* COVID.
- ▶ By 2024: grade 8 math back to its **2000** level; grade 8 and grade 12 reading the **lowest ever recorded**; cumulative decline $\approx 0.14\text{--}0.30$ SD ($\sim$ a full grade level at G8).
- ▶ 90–10 achievement gaps are the **widest in NAEP's history**.
- ▶ This talk: verify the facts from primary data, then **test every major explanation** against discriminating evidence — timing, distribution, cohorts, sectors, policy variation, international and adult parallels, and two causal designs.

All NAEP estimates pulled directly from the NCES Data Service API; replication repo with full audit trail.

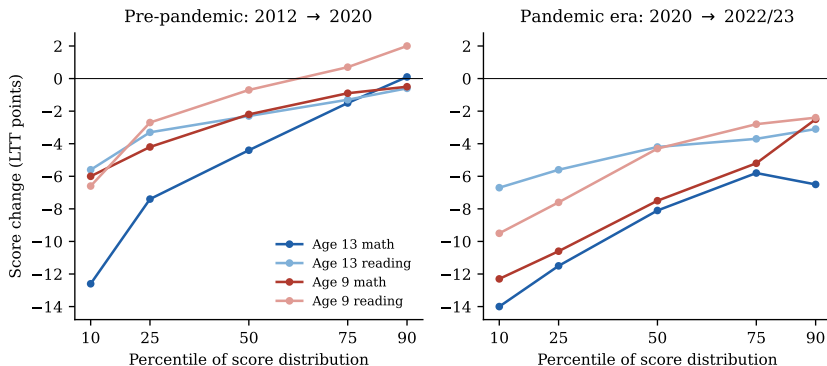
Preview of findings

1. The decline has **two phases with different causes**:
 - **2013–2019**: erosion confined to the *bottom* of the distribution;
 - **2019–present**: an unrecovered pandemic shock sustained by chronic absenteeism.
2. Direct tests **reject or marginalize** the standard school-policy suspects: demographics, funding, Common Core, NCLB-waiver deregulation, test effort, measurement artifacts.
3. The surviving phase-one explanation is **digital media saturation**: it alone fits the timing, the international and *adult* synchrony, the cohort structure, the sector neutrality, and the microdata exposure profile.
4. Two original causal exercises: a **waiver event study** (precise null) and the **first within-U.S. 4G-rollout design on test scores** (precise null on arrival timing — informative about mechanism).

Fact 1: every national series peaks ~ 2013 , two distinct drops



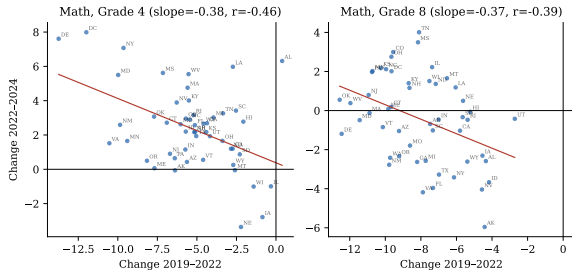
Fact 2: phase one is a collapse *at the bottom*



- ▶ Age-13 math, 2012→2020 (pre-pandemic): P10 -12.6 vs. P90 $+0.1$.
- ▶ 2022–24: the top recovers; **the bottom keeps falling**. G8 math 90–10 gap: 93 → 109.

NAEP Long-Term Trend; main-NAEP percentile series confirm (PC:P1–P9).

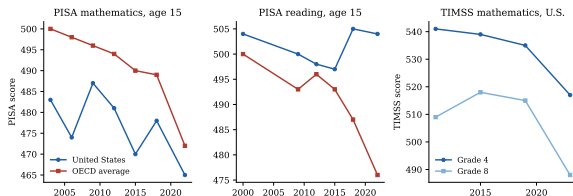
Fact 3: declines occur *within* every group, state, and sector



- ▶ Falling within every racial/ethnic and income group — composition explains ≤ 1 –2 points.
- ▶ Every state below its 2013 level in G8 math.
- ▶ Same trend in Catholic schools (more below).

NAEP main assessment, state panels and subgroup series.

Fact 4: not just American, and not just children



- ▶ PISA: synchronized pre-pandemic declines across rich countries with very different school policies.
- ▶ **Adults** (PIAAC): U.S. literacy **-12** 2017→2023; share at/below Level 1: 19%→28%; top stable. Declines in 19 OECD countries.
- ▶ Same bottom-heavy signature everywhere.

OECD PISA 2022 Vol. I; PIAAC Cycle 2 (caveats: 28% U.S. response rate, tablet-only).

The facts any explanation must fit

1. Onset ~2013, well before COVID timing
2. Concentrated at the bottom decile; top flat-to-rising pre-2020 distribution
3. Within every demographic group and state not composition
4. Uniform 2019–22 drop; post-2022 recovery only at the top two phases
5. Present in Catholic schools not school policy?
6. Synchronized internationally not U.S. policy
7. Extends to adults not schooling at all?
8. Accompanied by collapsing voluntary reading: 27% (2012) → 14% (2023)
mechanism clue

Eight candidate explanations

Schooling-side

- H1 Pandemic disruption
- H3 Retreat from NCLB accountability
- H4 Great-Recession funding cuts
- H8 Teacher quality / grade inflation

Student/society-side

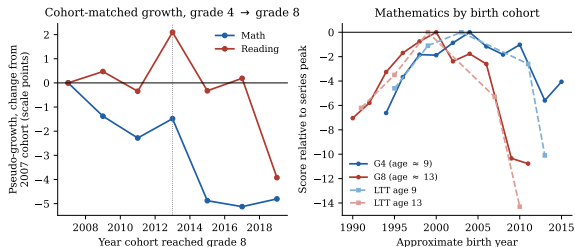
- H2 Smartphones & digital media
- H5 Demographic change
- H6 Chronic absenteeism
- H7 Declining reading practice

Strategy: derive *distinct predictions* from each across timing, distribution, cohorts, sectors, geography, international scope — then score all hypotheses against all facts.

Quick kills: demographics, funding, artifacts, effort

- ▶ **Demographics (H5):** declines occur within every group; compositional shift explains $\leq 1-2$ points of a 10+ point decline. **Rejected.**
- ▶ **Funding (H4):** spending recovered after 2013 while scores kept falling; credible \$-effects are an order of magnitude too small. **Rejected as a primary cause.**
- ▶ **Measurement artifacts:** exclusion flat at 1–3% (2013–2024); participation identical in 2022 and 2024. **Ruled out.**
- ▶ **Test effort:** PISA effort thermometer fell only 0.13–0.16/10 (2018→2022) $\Rightarrow$ $< 5\%$ of the decline; same decline on consequential state tests and among adults. **Bounded.**

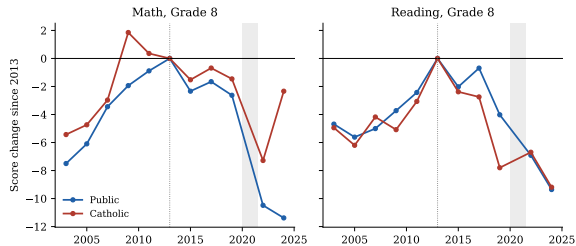
Test 1 — Cohorts: the losses accrue *during adolescence*



- ▶ Decompose every G8 change into **entry** (the cohort's G4 level) + **growth** (G4→G8).
- ▶ G8 reading 2013–19: $-4.4 = \text{entry} + 1.6 - \text{growth } 6.0$.
- ▶ Declining cohorts arrived at G4 at *record* levels, then lost ground in middle school: a **period effect in adolescence**, not a damaged-cohort story.

First formal cohort/period decomposition of the decline. Blind re-derivation reproduces it to the decimal.

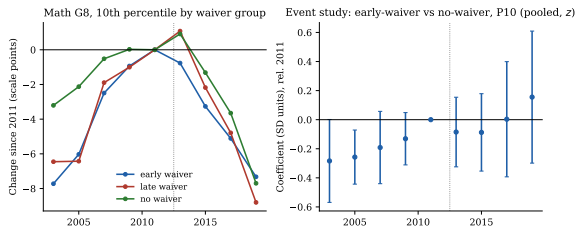
Test 2 — Sectors: Catholic schools declined in step



- ▶ Catholic schools were **never governed** by NCLB accountability.
- ▶ G8 reading 2013–19: Catholic -7.8 ($z = -3.4$) vs. public -4.0 — at least as large.
- ▶ Whatever drove phase one operated *across* the public/private boundary.

NAEP SCHTYPE series; published SEs. All-private reporting ends 2013, so Catholic is the usable benchmark.

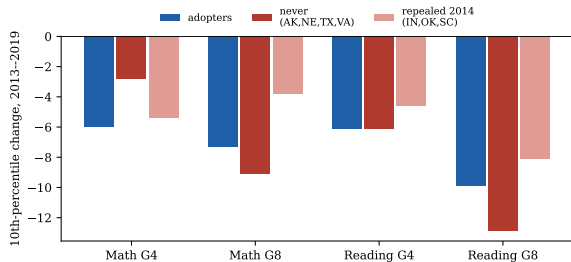
Test 3 — Policy variation: NCLB waivers (staggered event study)



- ▶ States exited NCLB accountability at **different times** (2012–15); seven never did.
- ▶ Dewey et al. (2026): this counterfactual has “never been tested.”
- ▶ Pooled DiD on state P10: ≈ 0 , **permutation** $p = 0.46$.
- ▶ Informative null: **MDE₈₀ ≈ 3.2 points** — under half the Dee–Jacob NCLB effect.
- ▶ Caveat: joint pre-trend $p = 0.08$.

Randomization inference, 2,000 permutations. Bleiberg (2020) binary/means version through 2013: also null.

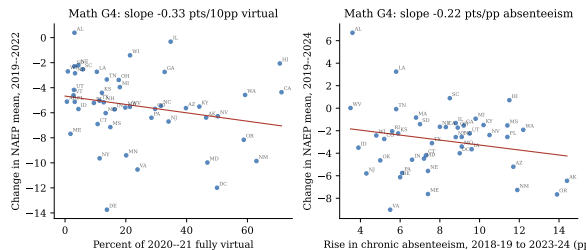
Test 4 — Common Core: never-adopters declined anyway



- ▶ AK, NE, TX, VA never adopted; their P10 declines are $\geq$ adopters' (diff -0.4 , $p = 0.72$).
- ▶ Minnesota (math-only non-adoption): its *non-CC* subject fell *less* — wrong sign for CC harm.

State adoption/repeal coding; triple-difference for MN.

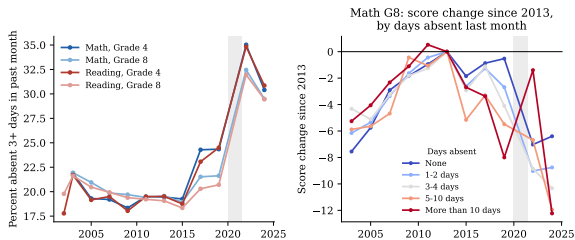
Phase two: the pandemic shock is real — but aggregation matters



- ▶ District-level closure effects are decisive (Goldhaber et al.; Jack-Oster).
- ▶ But **between states**, 2020-21 virtual share explains $\approx$ nothing of NAEP changes ($-0.04/10\text{pp}$, $p = 0.80$): state aggregation washes out district treatment variance.
- ▶ The puzzle is not the drop — it's the **non-recovery**.

CSDH schooling-mode shares $\times$ state NAEP changes; TUDA 26-district version similar (wide CIs).

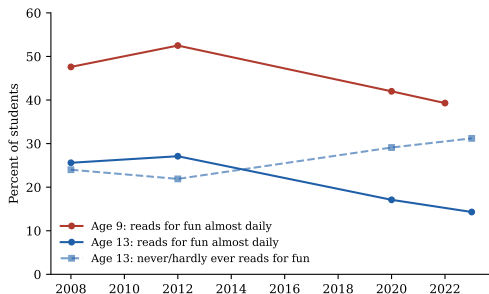
The recovery brake: absenteeism — with a pre-pandemic warning



- ▶ NAEP's own absence item: 3+ days missed last month rising *before* COVID (G4: 19.5% → 24% by 2019), then 32–35% in 2022, ~30% in 2024.
- ▶ Kitagawa: the absence shift accounts for **25–42% of 2019–24 declines** — but only **7–19% of pre-pandemic G8 declines**.
- ▶ Phase one was not an attendance phenomenon; phase two partly is.

B018101 crosstabs; decomposition is compositional, not causal. Blind re-derivation matches exactly.

The surviving hypothesis: digital media (H2) — timing and mechanism



- ▶ Teen smartphone ownership: 23% (2011) → 73% (2015) → 95% (2018). The peak-and-decline sits exactly on the takeoff.
- ▶ Reading for fun (age 13, almost daily): 27% → 14%, *mostly pre-pandemic*.
- ▶ H2 is the only hypothesis that natively explains the international *and adult* synchrony.

Pew; NAEP LTT student survey (S003501).

H2 in microdata: exposure sits exactly where the damage is

PISA 2022, 613,744 students; Rubin-combined PVs, school-clustered, ESCS-adjusted:

- ▶ Classroom device distraction (most/every lesson): **-13 (U.S.) / -15 (OECD)** points — survives SES adjustment essentially intact. 5+ hrs leisure device use: -40.
- ▶ Within countries, every over-engagement measure is **monotonically concentrated in the bottom score quartile** (distraction 32% vs. 24%; device anxiety 24% vs. 12%) ...
- ▶ ...yet **nearly flat across SES quartiles** (27.9% vs. 25.8%): a *low-performer* phenomenon, not a *low-income* one — matching the decline's signature in a way no compositional story does.
- ▶ Associational, two-way causality plausible; what it establishes is that **the exposure profile and the damage profile coincide**.

ST273Q06JA coding verified from file metadata (and independently re-derived blind). Item is rotated: U.S.

$n \approx 3,000$.

H2 causal corroboration from adjacent literatures

School phone bans (quasi-experimental):

- ▶ England: +0.064 SD, and +0.142 SD for low achievers, ≈ 0 for high — the mirror image of the decline (Beland–Murphy). Norway, Spain: similar; Florida 2023: gains by year two; Sweden's null where rules already bound.

Fertility (new 2025–26 rollout designs, same shock):

- ▶ iPhone access (AT&T-exclusivity design) cut teen births 4.5–8% (Myers–Hooper); 4G arrival (ruggedness IV) collapsed teen fertility, raised teen suicides (Hudson–Moscoso Boedo).
- ▶ Effects **monotone-declining in age, null for 25+**: the critics' objection to the baby-bust claim is **exactly our prediction** for an adolescent learning shock.

Cross-country: 3G arrival $\times$ PISA, 82 countries: -0.04 to -0.08 SD (Jain–Stemper).

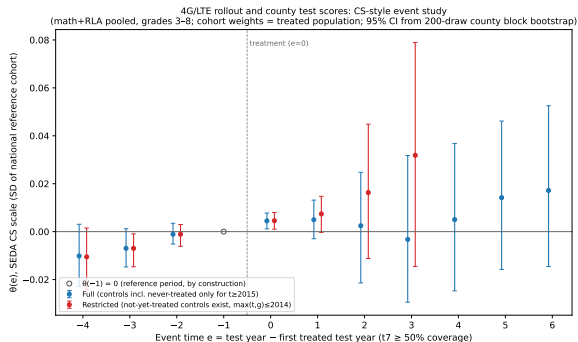
New test: the first within-U.S. mobile-rollout design on test scores

Design. Archived National Broadband Map: block $\times$ provider mobile coverage, 9 semiannual waves spanning the entire 2010–14 LTE rollout (validated against FCC benchmarks and Verizon's Dec-2010 launch markets). County treatment = first wave coverage $>50\%$ of population. Outcomes: SEDA 5.0 county means, grades 3–8, 2009–2019 (99.9% merge).

- ▶ Identifying variation = the **rural lag**: metro counties treated $\sim$ 2011–12, nonmetro 1–3 years later.
- ▶ Estimators: Callaway–Sant'Anna event study; continuous dose (county FE + state $\times$ year $\times$ grade $\times$ subject FE); within-county-year **adolescent-exposure gradient** across grades; permutation inference.

Wyckoff (2025): direct causal smartphone $\rightarrow$ achievement evidence “lacking.” Free data end to end; full pipeline in repo.

4G result: a precise null on the arrival-timing margin



- ▶ Dose: -0.002 **SD** (perm $p = 0.66$; **MDE**₈₀ = 0.013). Gradient: $+0.011/\text{yr}$ (perm $p = 0.12$; MDE 0.016).
- ▶ 10-spec robustness grid: all $|\hat{\beta}| \leq 0.003$, signs flip. Pre-trend drift disclosed.
- ▶ Reading: the decline did not ride on local cell-tower timing. The national, adoption-driven channel is invisible here (year FE; 3G-era adoption; controls gone by 2015).
- ▶ Contrast: the *same* variation moves teen fertility — not test scores.

Synthesis: a dual-criterion test

Borrowed from the fertility-puzzle debate (Kearney–Levine; Evans): a candidate cause must clear **two bars at once**:

1. **Synchrony:** explain the same decline across countries with disparate school systems — and among adults.
 2. **Signature:** explain bottom-of-distribution concentration, adolescent-period timing, sector neutrality.
- ▶ Every school-policy explanation **fails bar 1 by construction**.
 - ▶ The pandemic **fails bar 2** for phase one (it arrived six years late).
 - ▶ **Digital media is the only candidate examined that clears both.**
Accountability retreat retains a marginal supporting role (G4 public bottom decile); absenteeism owns the phase-two non-recovery.

Scorecard

	Timing	Bottom-heavy	COVID drop	No recovery	Within-grp	Int'l
H1 Pandemic	×	~	✓	~	✓	~
H2 Digital media	✓	~	×	✓	✓	✓
H3 Accountability	✓	✓	×	~	✓	×
H4 Funding cuts	~	×	×	×	~	×
H5 Demographics	×	×	×	×	×	×
H6 Absenteeism	×	✓	✓	✓	✓	~
H7 Reading practice	✓	~	×	✓	✓	✓
H8 Teachers/inflation	~	~	×	~	~	×

Two-phase accounting: phase one $\approx$ 20–45% of the total 2013–24 decline; pandemic window 30–70%; post-2022 the rest. **Full COVID recovery would still leave the U.S. far below 2013.**

What would settle it: a pre-registered test arriving in 2027

- ▶ ~ 30 states adopted school phone bans between 2023–24 and 2025–26, with staggered timing and varying bindingness (bell-to-bell vs. instructional-time).
- ▶ **Pre-registered design** (frozen before data exist): exposure groups from 50-state statute coding; NAEP 2026 state percentiles; same permutation machinery as the waiver test.
- ▶ Predictions on record: gains at P10/P25, larger under bell-to-bell bans; top decile $\approx$ flat.
- ▶ This is the waiver test's logic pointed at the leading hypothesis — with $\sim 4\times$ the treated states.

Also on the docket: restricted-use NAEP microdata; PISA 2012→2022 ICT panel; SEDA/adoption-margin designs.

Conclusion

1. The decline is real, large, verifiable — and **older than COVID**.
2. It is a crisis of the **struggling student**: bottom-decile collapse, record gaps.
3. Direct tests reject the school-policy suspects; the evidence triangulates to **digital media** for phase one and **absenteeism-sustained pandemic damage** for phase two.
4. Policy implications: treat attendance as the front line; take *binding* phone restrictions seriously (2026 will tell); target the bottom of the distribution; and do not mistake “back to 2019” for recovery — **2019 was already pointing down**.

Paper, data, code, claims audit, and pre-registration:

github.com/brendanbartanen-svg/naep-achievement-decline

Appendix: verification layer

- ▶ **Claims audit:** 41 load-bearing claims, each typed (citation/pulled/computed) with a <5-minute verification path; 62-assertion test suite runs clean.
- ▶ **Blind clean-room replications** (independent agents, no access to analysis code, fresh API/microdata pulls) — **5/5 reproduce**, no pipeline errors:
 - Cohort decomposition exact ($-4.45 = +1.57 - 6.02$); PISA distraction exact U.S. (-13.25), item coding independently recovered from file metadata; Kitagawa data to the decimal; Catholic-sector changes exact (official NAEP tests agree); waiver null + MDE confirmed, point estimate spec-dependent (≈ 0).
- ▶ **External anchors:** waiver null $\leftrightarrow$ Bleiberg (2020); state-aggregation null $\leftrightarrow$ Goldhaber/Jack-Oster; sector pattern $\leftrightarrow$ Malkus/Petrilli; PISA gaps $\leftrightarrow$ OECD published bivariate; 4G panel $\leftrightarrow$ FCC benchmarks + Verizon launch markets.
- ▶ **Human verification:** all four hand-coded policy datasets spot-checked against primary documents; five headline numbers re-pulled manually from the NAEP Data Explorer; Synthesis section adversarially reviewed by an independent colleague.
- ▶ Full record: report §Verification; evidence/claims_audit.md; verification/cleanroom/COMPARISON.md.

Appendix: 4G robustness grid

Specification	Coef. (SD)	p (cluster)
Dose, t7 (primary)	−0.0022	.57
Binary first_t7_50	−0.0017	.58
Binary first_t6_50	−0.0030	.35
Binary first_t7_25	−0.0022	.52
Binary first_t7_75	−0.0029	.28
Binary t6_50, Verizon-only	−0.0010	.81
Dose, drop artifact states	+0.0015	.64
Binary t7_50, drop artifact states	+0.0013	.63
Dose, unweighted	+0.0004	.92
Binary t7_50, unweighted	+0.0007	.81

County FE + state×year×grade×subject FE; SEs clustered by state; permutation inference within state.

Appendix: avenues examined and set aside

- ▶ **Opioid exposure:** state correlations null/inconsistent — tested and cut.
- ▶ **Cannabis legalization, discipline reform:** no clean variation — documented set-asides.
- ▶ **Cross-country smartphone-adoption timing:** data quality insufficient.
- ▶ **PISA leisure-hours items (ST326/ST322):** not administered in the U.S.; the IC171 fallback is an access artifact in a 95%-saturated population — cut.
- ▶ **TUDA district dose-response:** weak gradient, wide CIs — limitation, not contradiction, of district-level closure findings.